Capstone Design Project I

Progress Report



A Modular Self-Reconfigurable Robot (MSRR) for Search and Space Exploration

Submitted to the
Project Committee appointed by the
International School of Engineering (ISE)
Faculty of Engineering, Chulalongkorn University

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1/2026: 2147498 Capstone Design Project I
Robotics and Artificial Intelligence Engineering (International Program)
International School of Engineering (ISE), Faculty of Engineering,
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1 Project Timeline

- **Completed:** Project concept, objectives, literature review, and phased system architecture established.
- **In progress:** Phase A — single-module locomotion, sensing, communication, and simulation baseline.
- **Next:** Phase B — EP-Face docking, task-oriented reconfiguration, and simple assembled locomotion.

2 Project Phases

2.1 Phase A

External perception and computation. Develop the mechanically homogeneous module, differential-drive locomotion, joint control, odometry, and module-to-commander communication. Validate the module and physics twin before physical docking is introduced.

2.2 Phase B

Docking and task-oriented reconfiguration. Add the electropermanent EP-Face connectors, infrared alignment, and connector-state feedback. Extend the reconfiguration pipeline with task assignment, parallel assembly sequencing, and selection from a library of feasible configurations.

2.3 Phase C

Onboard perception with external computation. Mount a monocular RGB camera and an 8×8 multi-zone ToF sensor on the leader module. Estimate obstacle geometry and traversability locally, while transmitting sensor data to an external computer for processing.

2.4 Phase D

Fully onboard perception and computation. Move the perception and configuration-selection pipeline onto a coordinator module using an embedded compute board. Close the perceive–select–reconfigure–traverse loop on hardware without external localisation or computation.

3 Gantt Chart

Task schedule from the team Notion timeline (as of 20 Oct 2026), grouped by phase.

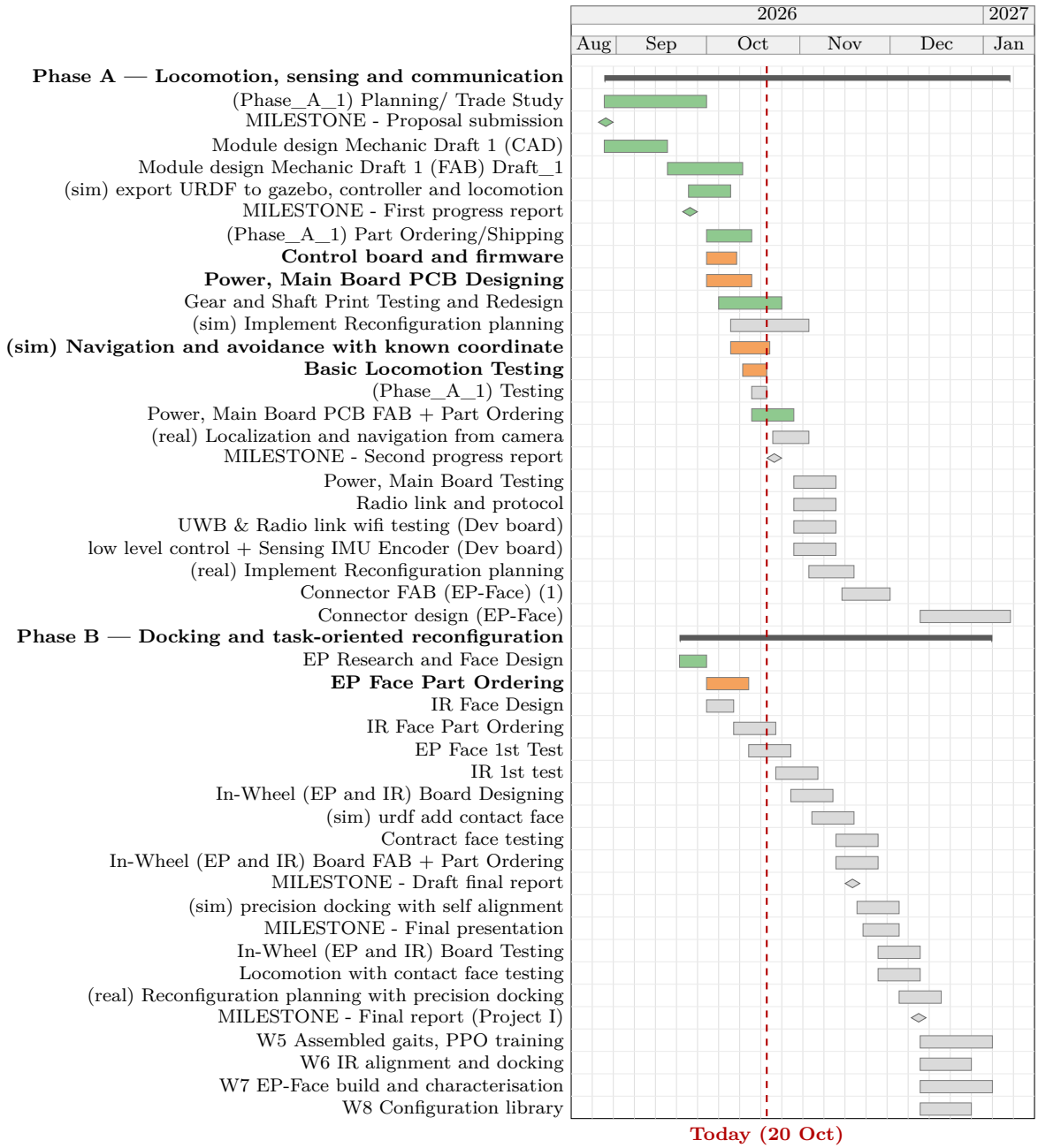


Figure 1: Project I schedule (Phases A–B). Completed In progress Upcoming; dashed line marks the report date.

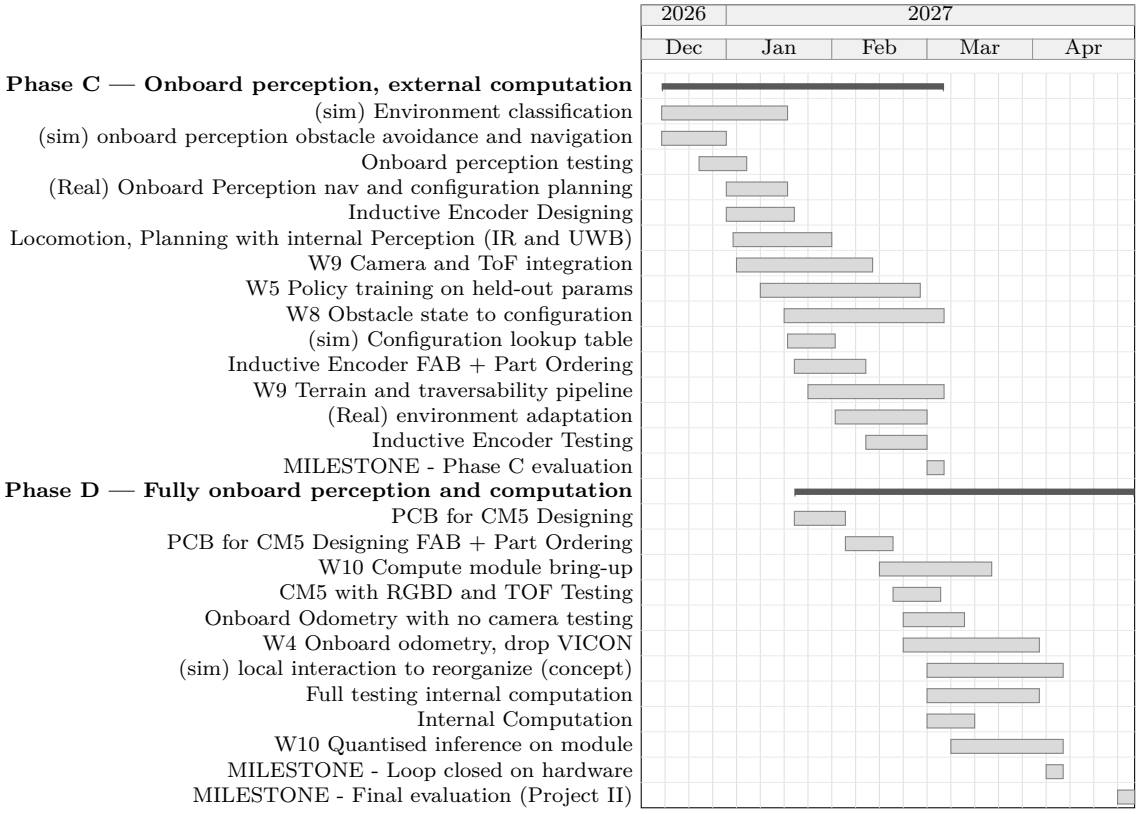


Figure 2: Project II schedule (Phases C–D). All tasks upcoming (▭).

4 Module CAD Design

The design is based on SMORES-EP [1] (originally 80 mm), scaled to a $100 \times 100 \times 100$ mm envelope for additional packaging space.

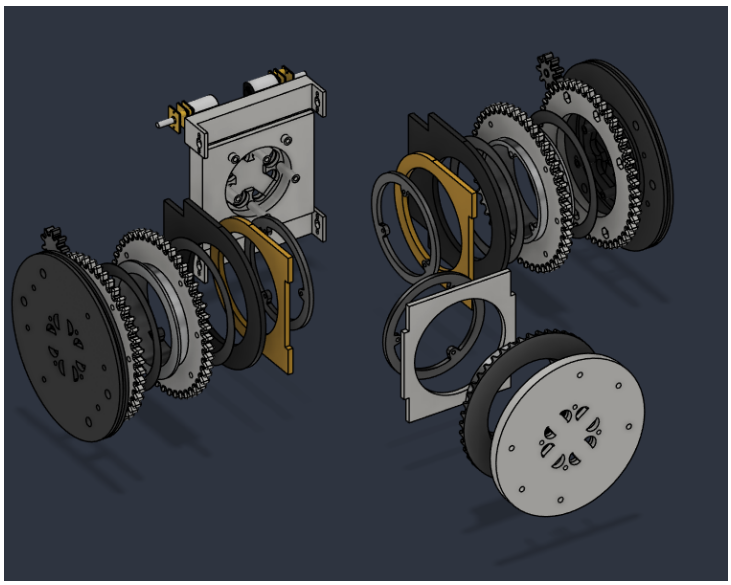
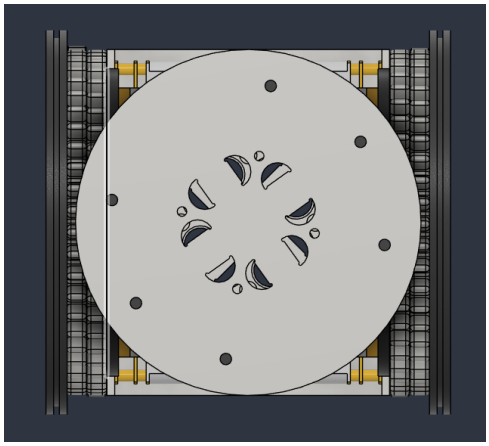
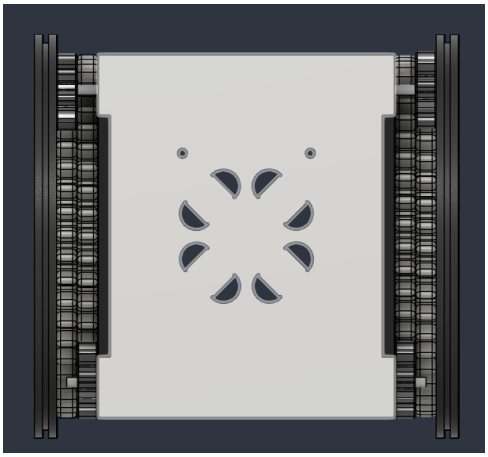


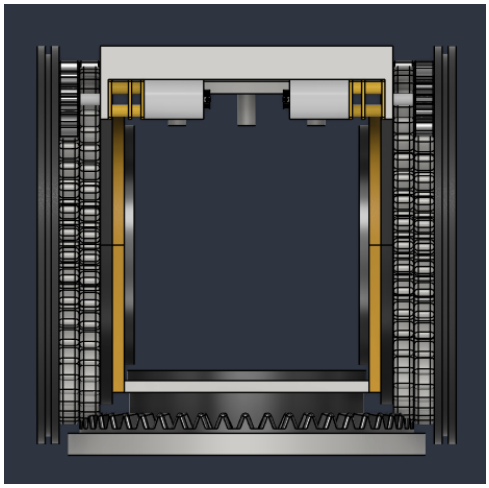
Figure 3: Exploded view of the module.



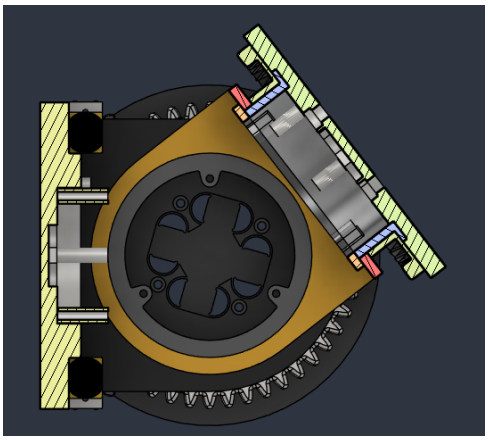
(a) Front view.



(b) Rear view.



(c) Top view.



(d) Tilted sectional view.

Figure 4: Module CAD views.

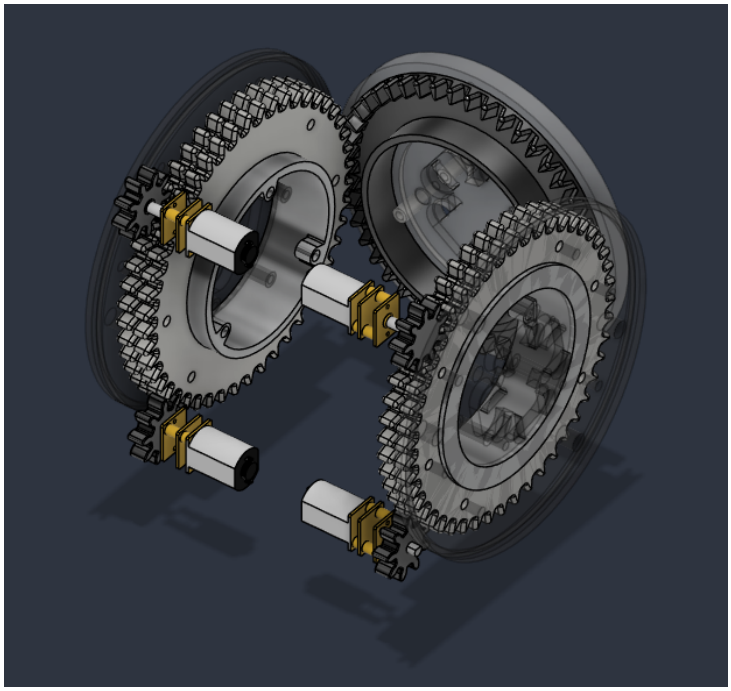


Figure 5: CAD view of the gear train and motor arrangement (48T spur gears, 9T pinions, 46T crown gear).

4.1 Gear Train Design

4.1.1 SMORES-EP Reference Baseline

The original SMORES-EP gear train was reverse-engineered from published data (Table 1). This gives module $m = 1.4$, a spur pitch diameter of $1.4 \times 48 = 67.2\,\text{mm}$ (OD $= 1.4 \times 50 = 70\,\text{mm}$), and a stage ratio of $48/9 \approx 5.33$.

Table 1: Reconstructed SMORES-EP reference parameters.

Parameter	**SMORES-EP**
Module envelope	$80 \times 80 \times 80\,\text{mm}$
Module m	1.4
Spur gear	48 teeth (PD 67.2 mm, OD 70 mm)
Pinion	9 teeth
Stage ratio	$48/9 \approx 5.33$:1
Crown gear	Same part as spur, used at 90°
Gearbox	298:1

4.1.2 Spur Gear and Pinion

After wall thickness, bearing seats, and running clearance, the 100 mm envelope allows a maximum spur gear OD of 90 mm. The final design keeps the SMORES-EP tooth counts (48T spur, 9T pinion) and scales the module from $m = 1.4$ to $m = 1.8$, so the spur OD

is $m(z + 2) = 1.8 \times 50 = 90\,\text{mm}$ and fills the available diameter exactly. The resulting spur–pinion centre distance is $m(z_1 + z_2)/2 = 1.8 \times 57/2 = 51.3\,\text{mm}$.

Reasons for this choice:

- **Ratio preserved.** The $48/9 \approx 5.33{:}1$ stage ratio is unchanged, so the gearbox selection, the SMORES-EP performance benchmarks, and the control/firmware tuning all remain valid.
- **Direct comparability.** Using the same tooth counts as SMORES-EP means the gear train differs from the reference only in scale.
- **Larger, stronger teeth.** At $m = 1.8$ the tooth thickness on the pitch circle is $\pi m/2 \approx 2.83\,\text{mm}$, compared with 2.14 mm for $m = 1.36$. This makes the teeth stronger in FDM nylon and less sensitive to print tolerances.
- **Adequate pinion size.** The 9T pinion has an OD of $1.8 \times 11 = 19.8\,\text{mm}$, about the same as the 12T alternative (19.04 mm), so there is still enough material around the motor-shaft bore.

Design change from the previous draft. An earlier draft proposed a 64T/12T pair at $m = 1.36$ (same 5.33:1 ratio) to reduce pinion undercut and raise the contact ratio. This was replaced by the 48T/9T, $m = 1.8$ pair, which meshed and turned smoothly in print testing.

Pinion undercut. A 9-tooth pinion at a 20° pressure angle is well below the theoretical minimum tooth count for no undercut ($z_{\min} = 2/\sin^2\alpha \approx 17$), so some root undercut is unavoidable. SMORES-EP has the same undercut. It was accepted because the printed pair meshes and runs smoothly.

Motor-shaft D-bore. The pinions mount on the N20 gearmotor D-shaft. The bore was first modelled with a 0.1 mm tolerance. To allow for shrinkage of printed nylon holes, the D-bore tolerance has since been made slightly larger. The revised bore has not yet been print tested.

4.1.3 Pan/Tilt Crown Gear

SMORES-EP reuses a spur gear as a crown gear at 90°. This design instead uses a **true crown (face) gear** to improve load distribution across the mesh. A first attempt at 90 mm OD interfered with surrounding components (motors, gearboxes, and structural frame), so the crown gear was reduced to 84 mm OD with 46 teeth and $m = 1.75$ (pitch diameter 80.5 mm).

Tooth proportions. The crown addendum is the standard $h_a = m = 1.75\,\text{mm}$. The dedendum does *not* use the standard $h_f = 1.25m = 2.1875\,\text{mm}$; it is set to 2.15 mm instead. With the 1.8 mm addendum of the mating spur gear, this leaves a root clearance of 0.35 mm.

Module mismatch. The spur gears ($m = 1.8$) and the crown gear ($m = 1.75$) have different modules. This is an accepted compromise, because the spur diameter, crown diameter, and tooth counts are all fixed by packaging and ratio constraints. The mismatch is small (about 2.8 %). The circular pitch is $\pi m = 5.65\,\text{mm}$ for the spur gear and 5.50 mm for the crown gear, a difference of about 0.16 mm per tooth, which is the same order as

the 0.2 mm backlash allowance. Print testing confirmed that the spur–crown mesh turns smoothly despite the mismatch.

4.1.4 Final Gear Parameters

Table 2: Final gear train parameters.

Component	Teeth	Module	PD	OD	Notes
Spur gear (×4)	48	1.8	86.4 mm	90 mm	5 mm face width, bolt holes in web
Pinion (×4)	9	1.8	16.2 mm	19.8 mm	D-bore tolerance slightly enlarged (not yet print tested); through-bore (pan/tilt) or blind (wheel)
Crown gear	46	1.75	80.5 mm	84 mm	True face gear; $h_a = 1.75$ mm, $h_f = 2.15$ mm

Common to all gears: 20° pressure angle, 0.2 mm backlash, and 0.3 mm root fillet. The backlash and fillet are deliberate manufacturing allowances. The spur–pinion centre distance is 51.3 mm. All gears were generated with the Fusion 360 spur gear tool and 3D-printed in nylon.

4.1.5 Drivetrain Architecture

Four motors each drive a 9T pinion meshing with a 48T spur gear:

- **Outer two:** each drives a wheel (differential-drive locomotion).
- **Inner two:** both mesh with a single crown gear, which works as a differential:
  - Both spur gears turning the *same* direction about their shared axis → the crown gear does not spin; the whole crown assembly rotates about the shared axis (**tilt**).
  - Spur gears turning in *opposite* directions → the crown gear spins about its own axis (**pan**).

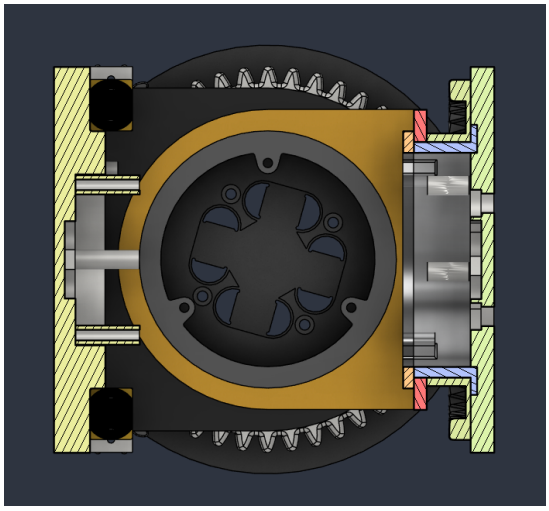
The crown gear must mesh symmetrically with both inner spur gears. Any asymmetry would couple the pan and tilt motions.

4.1.6 Status

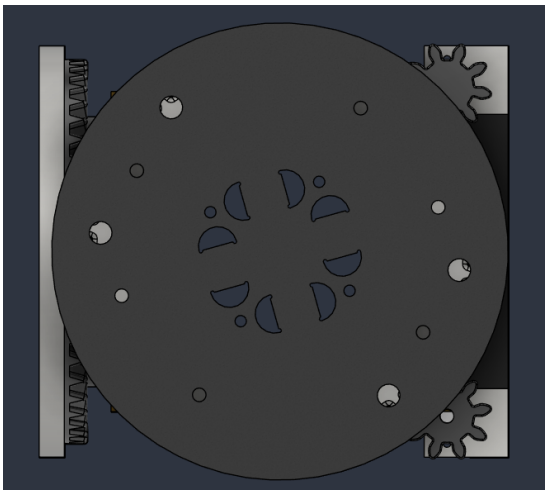
All gears (48T spur, 9T pinion, and 46T crown) have been printed and tested. Both the spur–pinion and spur–crown meshes turn smoothly. The remaining step is to print test the revised pinion D-bore on the motor shaft.

4.2 Outer Wheels

Two independently driven wheels (100 mm diameter) with NBR N70 O-ring tire groove (1.5 mm cross-section), EP-magnet docking space, and PCB mounting posts.



(a) Interior view (wheel and drivetrain).



(b) Exterior view (wheel and module).

Figure 6: Outer wheel CAD.

4.3 Pan Face

90 mm diameter, no O-ring groove. Includes EP-magnet docking space.

4.4 Tilt Face

Approximately 90×71.6 mm rectangular part with EP-magnet docking space and an additional motor mount.

4.5 Simulation Model

The module CAD is exported from Blender (LinkForge) to URDF and refactored into xacro, then simulated in Gazebo Fortress with `ros2_control` on ROS 2 Humble. The model serves as a digital twin for testing control algorithms before hardware is available.

4.5.1 Kinematic Model

The model has five links (base, two wheels, tilt body, pan face) and four actuated joints (Table 3). The exporter placed the CAD orientation of each part in the joint frames, giving rotated and inconsistent TF frames. All link frames were therefore re-aligned with the base following REP-103 (x forward, y left, z up), and each mesh's CAD rotation was moved into its visual and collision origin. A forward-kinematics comparison at random joint configurations confirmed every mesh pose was unchanged (maximum difference 1×10^{-6} m). Self-collision between the module's own links is disabled.

Table 3: Simulated joints and control interfaces.

Joint	Type	Axis	Command	Controller
`left_wheel_joint`	continuous	$+y$	velocity	`diff_drive_controller`
`right_wheel_joint`	continuous	$+y$	velocity	`diff_drive_controller`
`tilt_body_joint`	revolute, $\pm 180°$	$-y$	position	`head_position_controller`
`pan_face_joint`	continuous	$-x$	position	`head_position_controller`

The pan/tilt differential is not modelled mechanically. The simulation exposes pan and tilt directly as joints, and the mixing to the two inner motors is left to the firmware:

$$\theta_A = N_t\,\theta_{\text{tilt}} + N_p\,\theta_{\text{pan}}, \qquad \theta_B = N_t\,\theta_{\text{tilt}} - N_p\,\theta_{\text{pan}}, \tag{1}$$

where θ_A, θ_B are the inner motor angles and N_t, N_p the motor-to-joint reductions of the gear train. As a consequence, the simulation does not reproduce the shared speed and torque limit when pan and tilt move together. For the first milestone, tilt and pan are held at 0 rad.

4.5.2 Inertial Parameters

The exporter's inertial values were placeholders. Every link had $I_{xx} = I_{yy} = I_{zz} = m/6$, which is the inertia of a solid 1 m cube ($I = m(a^2 + a^2)/12$ with $a = 1\,\text{m}$), regardless of shape. For parts about 0.1 m across this overestimates inertia by roughly 100×, and the centres of mass were the mesh offsets rather than true centroids.

The values were therefore recomputed from the CAD meshes, assuming a uniform solid of 3D-printed nylon PA12, $\rho = 1010\,\text{kg/m}^3$. Each closed triangle mesh is decomposed into tetrahedra formed by every face $(\mathbf{a}_k, \mathbf{b}_k, \mathbf{c}_k)$ and the origin [**?**]:

$$V_k = \tfrac{1}{6}\,\mathbf{a}_k \cdot (\mathbf{b}_k \times \mathbf{c}_k), \qquad m = \rho \sum_k V_k, \qquad \mathbf{c} = \frac{1}{\sum_k V_k} \sum_k V_k\,\frac{\mathbf{a}_k + \mathbf{b}_k + \mathbf{c}_k}{4}, \tag{2}$$

$$C = \rho \sum_k \frac{V_k}{20}\left[\mathbf{a}_k\mathbf{a}_k^T + \mathbf{b}_k\mathbf{b}_k^T + \mathbf{c}_k\mathbf{c}_k^T + \mathbf{s}_k\mathbf{s}_k^T\right], \qquad \mathbf{s}_k = \mathbf{a}_k + \mathbf{b}_k + \mathbf{c}_k, \tag{3}$$

$$I_{cm} = \operatorname{tr}(C)\,\mathbf{E}_3 - C - m\left[(\mathbf{c}^T\mathbf{c})\,\mathbf{E}_3 - \mathbf{c}\mathbf{c}^T\right], \qquad I_{\text{link}} = R\,I_{cm}\,R^T, \tag{4}$$

where the signed volumes V_k cancel outside the solid, C is the second-moment matrix, $\mathbf{E}_3$ is the 3×3 identity, the bracketed term is the parallel-axis shift to the centre of mass, and R rotates the result into the link frame. The method is exact for a polyhedron. Implemented in `compute_inertia.py`, it reproduced the analytic solution of an $80 \times 50 \times 30$ mm box with 0.000 % error and of a 256-facet cylinder ($r = 35$ mm, $h = 12$ mm) within 0.02 %, the residual being faceting. Results are listed in Table 4; the wheels give $I_{yy} \approx 2I_{xx}$, as expected for a disc spinning about y.

Table 4: Computed mass properties (solid nylon PA12, inertia about the centre of mass).

Link	**Volume (cm^3)**	**Mass (g)**	I_{xx} **(kg m^2)**	I_{yy} **(kg m^2)**	I_{zz} **(kg m^2)**
Base	139.95	141.3	2.51×10^{-4}	2.05×10^{-4}	2.57×10^{-4}
Left wheel	51.15	51.7	3.55×10^{-5}	7.00×10^{-5}	3.55×10^{-5}
Right wheel	51.15	51.7	3.57×10^{-5}	7.00×10^{-5}	3.53×10^{-5}
Tilt body	28.39	28.7	3.46×10^{-5}	2.82×10^{-5}	3.33×10^{-5}
Pan face	37.11	37.5	4.36×10^{-5}	2.21×10^{-5}	2.22×10^{-5}

The main remaining uncertainty is in the inputs, not the method: material density, partial infill if the parts are printed by FDM, and components absent from the mesh (motors, battery, PCBs), which make the base too light. Measured part masses will replace the density estimate once parts are printed; the script accepts them directly and rescales the inertia.

4.5.3 Multi-Module Simulation

N identical modules are spawned (default three), each in its own namespace (`module_1`, ...) with its own controller manager, controllers and TF frame prefix, so modules can be commanded independently. The wheels use a differential-drive controller (wheel separation 96.65 mm, radius 50 mm, both measured from the meshes) driven by velocity commands (v, ω).

The first milestone localises modules with AprilTags. To test algorithms before the vision pipeline exists, each module's exact pose is taken from the physics engine and published as `/module_i/ground_truth/odom`. A separate package, `msrr_control`, reads these poses and commands all modules from one node. It is limited to the real top speed of 0.049 m/s (50 rpm motor output through the 5.33:1 wheel gear). The AprilTag estimate will later replace the ground-truth topic without changing the algorithm code. A go-to-goal test with three modules reached all goals within 1 cm.

4.5.4 Status and Limitations

- Joint effort and velocity limits are placeholders, pending actuator characterisation.
- The pan/tilt motor coupling and gear backlash are not simulated.
- Docking is not modelled; Phase B will use Gazebo's detachable joints to emulate the EP-Face connection.
- Each module is supported only by its wheels and settles with a 4° to 6° lean; contact with the ground will be checked against the printed module.

5 Electronics

5.1 Phased Electronics Plan

Electronics are built incrementally across project phases:

- **Phase A-1:** COTS parts — 2S battery, 4-channel motor driver with MCU, ESP32-C3 for Wi-Fi.
- **Phase A:** Custom PCBs — power board (buck, LDO, 4× H-bridge with current sensing) and main board (STM32G474, IMU, UWB, Wi-Fi).
- **Phase B:** In-wheel PCB with EP-Face circuitry and IR contact-alignment, connected via slip ring.
- **Phase C:** RGB-D/ToF camera (Sipeed MaixSense A075V) + wheel encoder added; compute still external.
- **Phase D:** Raspberry Pi CM5 on custom carrier PCB moves compute onboard.

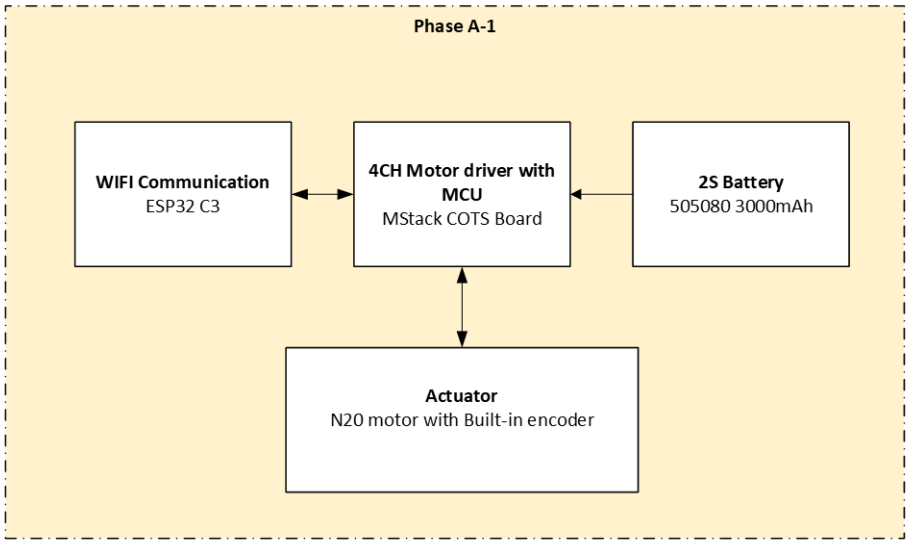
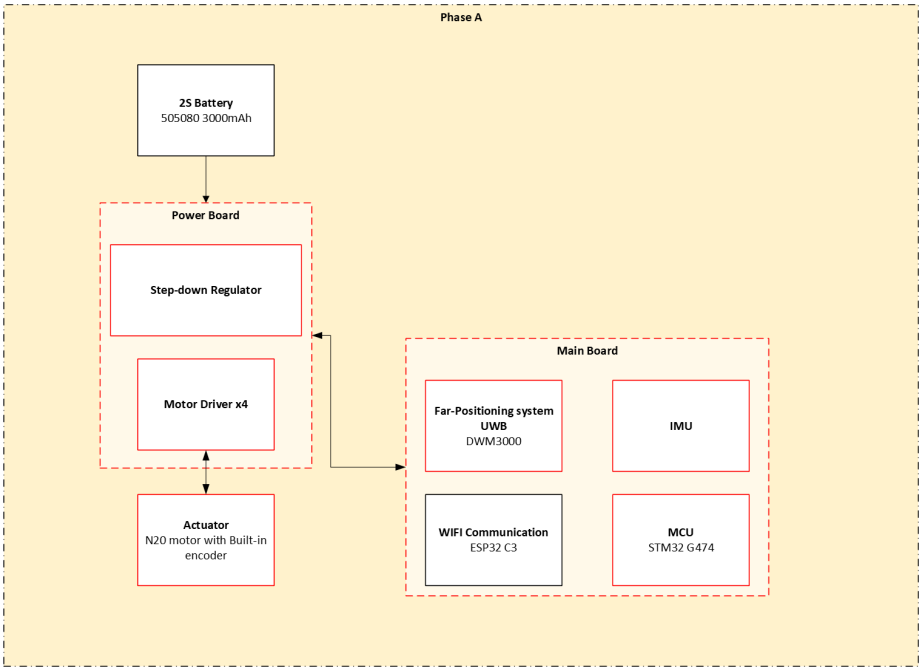
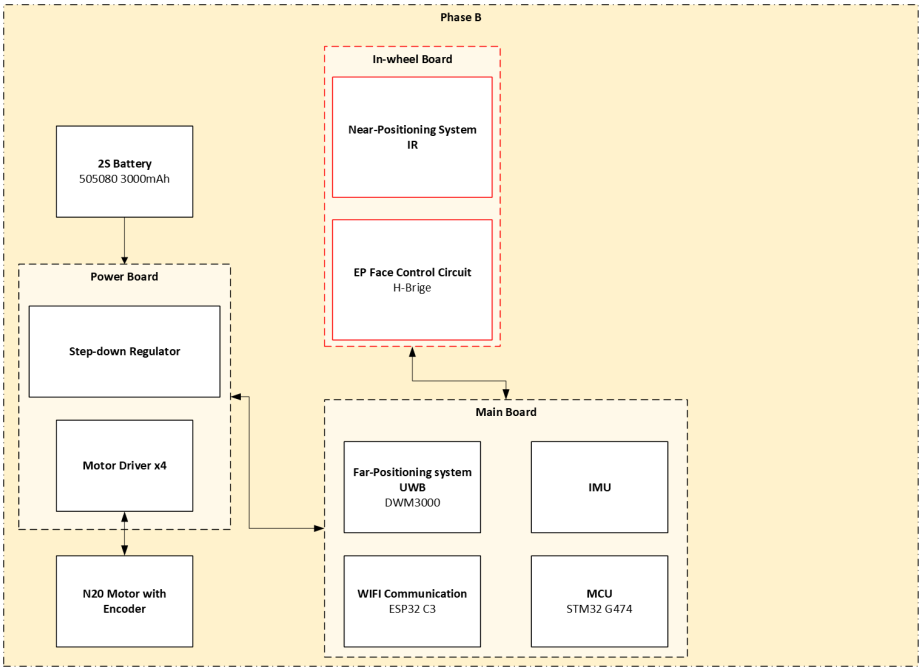


Figure 7: Phase A-1 electronics block diagram.

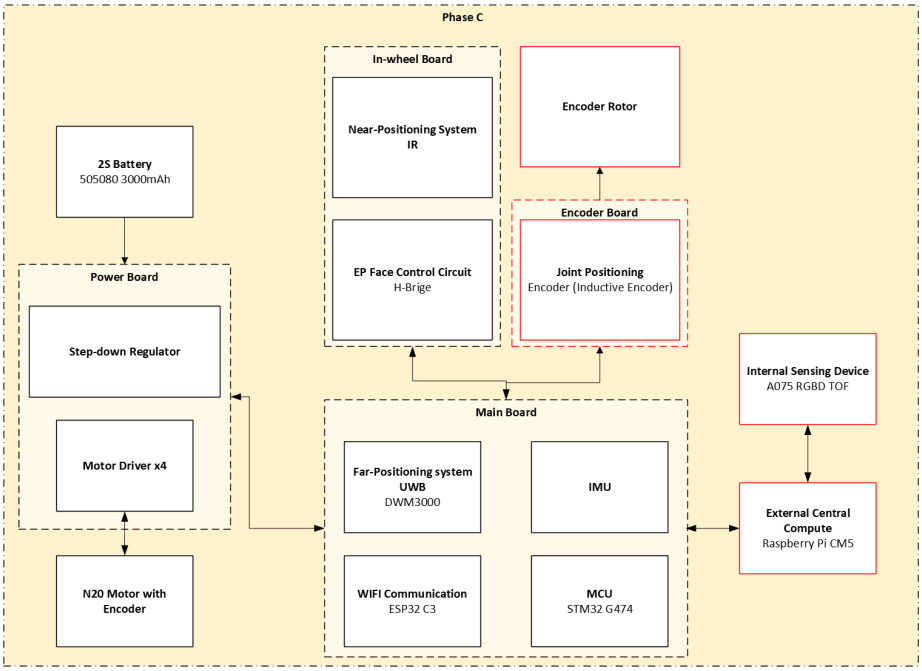


(a) Phase A.

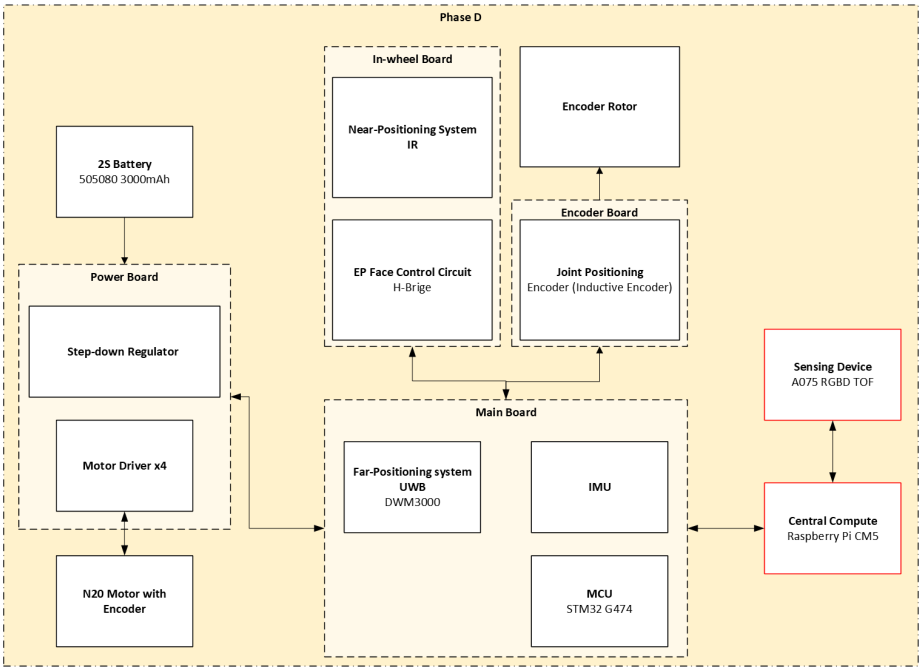


(b) Phase B.

Figure 8: Electronics block diagrams, Phases A–B.



(a) Phase C.



(b) Phase D.

Figure 9: Electronics block diagrams, Phases C–D.

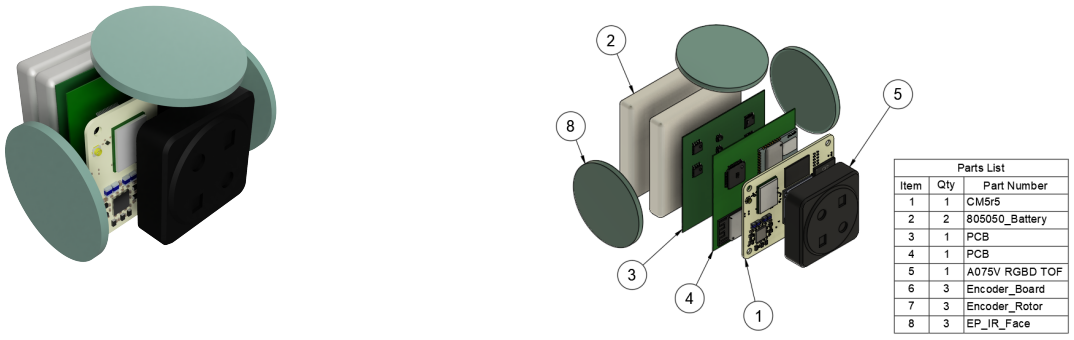
5.2 Component Selection

Table 5: Key electronics components.

Component	Phase	Key Specs	Interface / Supply
4× N20 gearmotor	A-1+	6 V, 50 rpm; 7 PPR encoder	A/B quadrature; H-bridge
STM32G474	A+	Cortex-M4F @ 170 MHz; 12-bit ADC; FDCAN	SPI/I2C/UART/CAN; 3.3 V
ICM-42688-P	A+	±2000 dps / ±16 g; 32 kHz ODR	SPI or I2C; 1.8 V
DW3000 (UWB)	A+	IEEE 802.15.4z; ∼10 cm ranging	SPI; 3.3 V
ESP32-C3	A-1+	Wi-Fi 4 + BLE 5; RISC-V @ 160 MHz	UART/SPI; 3.3 V
EP-Face + IR PCB	B	Custom; EP magnet + IR alignment	Slip ring; 3.3 V
RPi CM5	D	Cortex-A76 @ 2.4 GHz; up to 16 GB RAM	Custom carrier; 5 V
Sipeed MaixSense A075V	C+	ToF 320×240 @ 30 fps; RGB 800×600	USB 2.0; 5 V

5.3 Electronics Packaging

All components fit within the 100 mm cube based on rough CAD placement (Figure 10).



(a) Assembled. **(b)** Exploded.

Figure 10: Electronics packaging.

5.4 Wheel Encoder

Replacing the original continuous-rotation potentiometer (∼5° resolution, contact wear) with a non-contact encoder. Three options considered:

Table 6: Wheel encoder options.

	On-axis magnetic (AS5600)	**Off-axis tape**	**Custom inductive**
Output	Absolute, 12–14 bit	Incremental + index	Absolute
EP interference	Sensitive	Sensitive	Immune
Design effort	Low	Medium	High

The inductive option best handles the EP connector's nearby field but has the highest schedule risk. Decision deferred to Phase C.

5.5 Changes from Progress Report 1

The phased plan (A-1 to D) and its scope are unchanged. The electronics work plan is delayed by one week, which shifts the electronics timeline and the work that depends on it by the same amount.

Phase A-1 was built from an STM32F411 development board, two TB6612FNG dual H-bridge drivers and an ESP32-C6 module, instead of the COTS 4-channel motor driver with a built-in MCU and the ESP32-C3 described in PR1. This gives direct control of the PWM, direction and encoder signals, so the wiring and firmware carry over to the Phase A main board.

The only other hardware change is the battery. The 2S 3000 mAh 505080 pack was replaced with a 3S pack of Vapcell M11 18350 Li-ion cells (1100 mAh, 10 A continuous, 15 A burst), which can supply more current for the EP-Face in Phase B. The smaller capacity lowers the estimated run time (Section 5.7).

5.6 Phase A-1 Hardware

Phase A-1 is the bench version of the locomotion electronics. It tests the wiring, power path and firmware set-up that will move onto the Phase A PCBs.

5.6.1 Hardware Schematic

Figure 11 shows the Phase A-1 wiring. An STM32F411CEU6 development board controls four N20 gearmotors through two dual-channel TB6612FNG H-bridge drivers (two motors each, shared standby line). Each motor has a PWM input and two direction inputs, and its quadrature encoder is read back by the MCU. M1 and M2 drive the left and right wheels; M3 and M4, the inner-left (IL) and inner-right (IR) motors, drive pan and tilt. An ESP32-C6 module provides the Wi-Fi link over UART, an ST-Link handles SWD programming, and USB gives a serial console for testing.

Three buck converters run from the 3S pack through an XT30 connector. A 5 V buck supplies the MCU board and the ESP32-C6; the MCU board's 3.3 V regulator supplies the driver logic and the encoders. One 6 V buck per TB6612FNG supplies the motors, so the N20s run at their rated voltage at full PWM duty. A 100 nF capacitor across each motor suppresses brush noise, and a 10 kΩ pull-down on STBY keeps both drivers disabled until the MCU enables them.

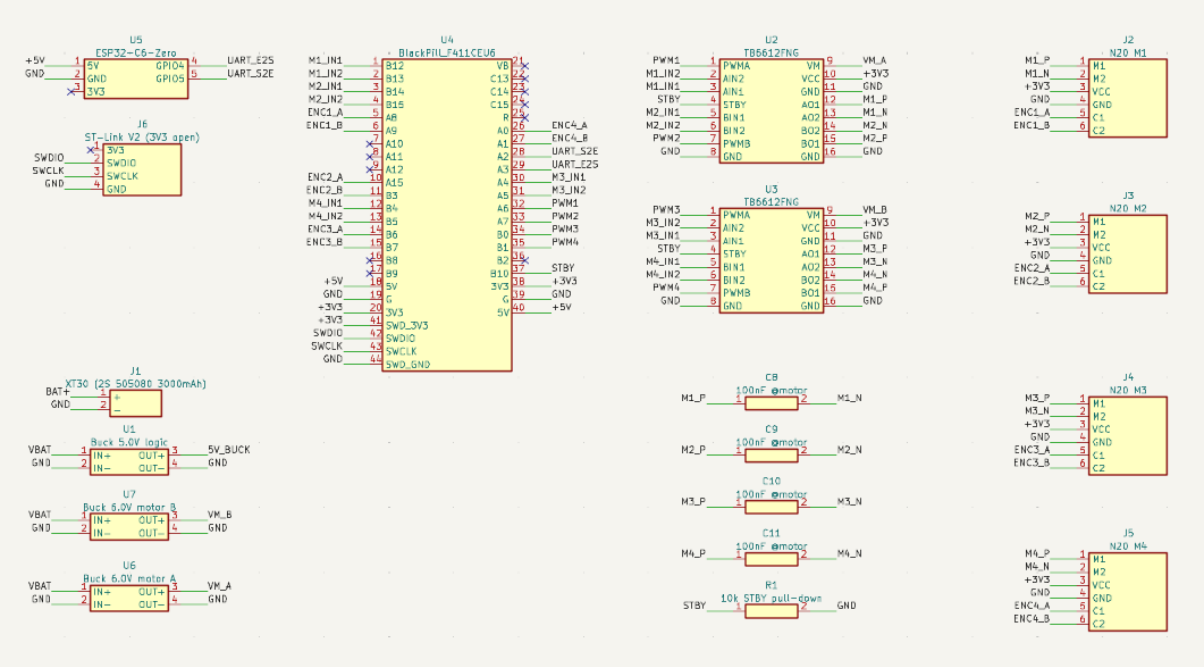
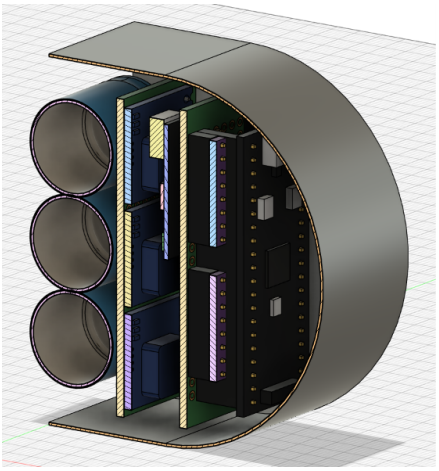


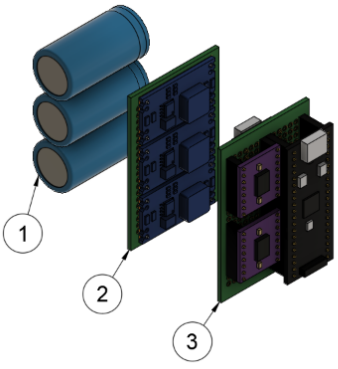
Figure 11: Phase A-1 hardware schematic.

5.6.2 Hardware Packaging

Figure 12 shows the Phase A-1 hardware placed in the module. Three parts stack front to back inside the module shell. The three 18350 cells sit in a vertical column at the back. In front of them is the regulator board with the three buck converters. The controller board, carrying the STM32F411 board, both TB6612FNG drivers and the ESP32-C6, sits at the front.



(a) Assembled.



(b) Exploded.

Parts List			
Item	Qty	Part Number	Material
1	1	Battery_Pack	Steel
2	1	MSRR_A1_board2_regulators	Steel
3	1	MSRR_A1_board1_esp_flat	Steel

Figure 12: Phase A-1 hardware packaging.

5.6.3 Power Calculation

Table 7 gives the Phase A-1 load on the 3S pack, using the PR1 load figures. Motor power passes through the 6 V regulators (90 %) and the TB6612FNG (95 %); logic power through a 5 V buck (90 %) and the board's 3.3 V LDO (66 %).

Table 7: Phase A-1 power budget (3S pack, 9.8 Wh usable).

Load	Typical	All motors driving	Peak (all stalled)
4× N20 via 6 V reg. + TB6612	2.6 W[a]	7.0 W	28.1 W
STM32F411 + ESP32-C6 (3.3 V)	0.7 W	0.7 W	2.2 W[b]
Total from pack	**3.3 W**	**7.7 W**	**30.3 W**
Pack current (11.1 V)	0.30 A	0.69 A	2.7 A (3.4 A at 9 V)
Run time	**3.0 h**	**1.3 h**	–

[a]PR1 duty cycle: wheels 60 %, pan/tilt 15 %, 0.25 A each loaded. [b]Wi-Fi transmit burst, 335 mA (PR1 ESP32-C3 figure).

The 3.4 A worst-case pack current is well inside the cell's 10 A continuous rating. Because the drivers see 6 V, a stalled N20 draws about 1 A, inside the TB6612FNG's 1.2 A continuous rating per channel. Each 6 V regulator must supply up to 2 A with both of its motors stalled.

5.6.4 Embedded System Setup

The STM32F411 was configured in STM32CubeMX (Figure 13, Table 8). The 25 MHz crystal is multiplied by the PLL to a 96 MHz system clock, which also gives the exact 48 MHz needed for USB. One timer generates the PWM for all four motors at 20 kHz, which is above hearing range. Each encoder has its own timer in quadrature (4×) mode with input filtering and pull-ups. A separate timer interrupt at 1 kHz sets the control-loop rate. The ESP32-C6 link uses DMA, with a circular receive buffer so incoming commands are not lost while the CPU is busy.

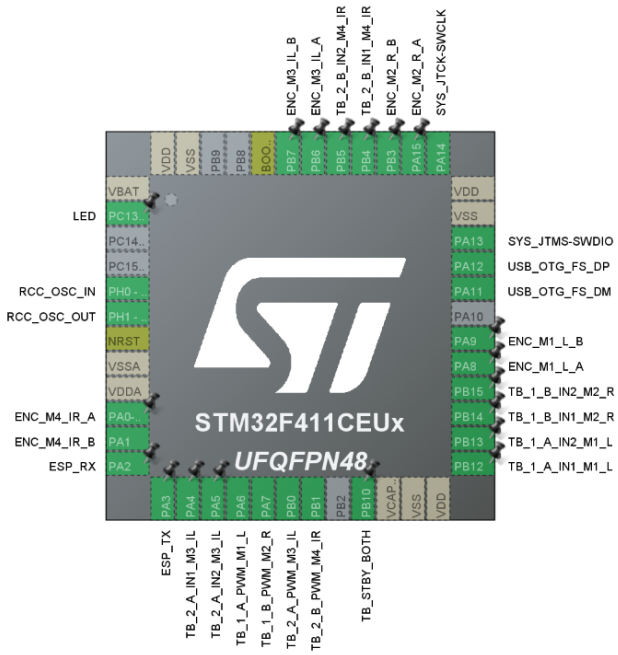


Figure 13: STM32CubeMX pin and peripheral configuration.

Table 8: STM32F411 peripheral assignment.

Function	Peripheral and pins	Setting
System clock	HSE 25 MHz → PLL	96 MHz
Motor PWM, M1–M4	TIM3 CH1–4 (PA6, PA7, PB0, PB1)	20 kHz
Motor direction	GPIO, 2 per motor (PB12–15, PA4–5, PB4–5)	–
Driver standby	GPIO PB10 (both TB6612)	–
Encoder M1 (L)	TIM1 (PA8, PA9), 16-bit	×4 quadrature
Encoder M2 (R)	TIM2 (PA15, PB3), 32-bit	×4 quadrature
Encoder M3 (IL)	TIM4 (PB6, PB7), 16-bit	×4 quadrature
Encoder M4 (IR)	TIM5 (PA0, PA1), 32-bit	×4 quadrature
Control loop tick	TIM10 interrupt	1 kHz
ESP32-C6 link	USART2 (PA2, PA3) + DMA	115 200 baud
PC console	USB OTG FS, CDC class	–
Debug	SWD (PA13, PA14)	–

5.7 New Power Calculation

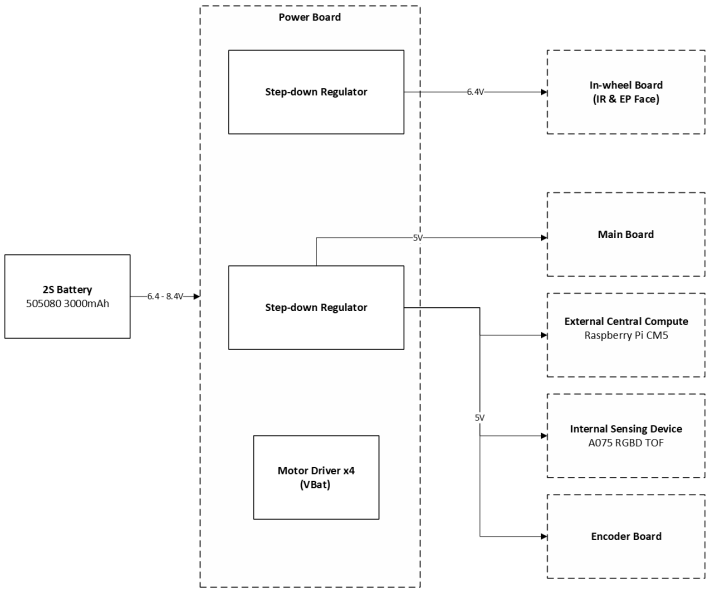


Figure 14: Power distribution block diagram.

The PR1 load budget is unchanged; only the pack changes. The Phase A power board keeps the PR1 architecture: the H-bridges run directly from the pack, not from the 6 V regulator used on the Phase A-1 bench set-up. This removes the regulator loss and a 4 A regulator from the board. Power from the 3S pack splits three ways:

- **Motors:** 4× N20 via H-bridges, PWM capped at ∼48 % (6/12.6 V at full charge) for the 6 V rating.
- **EP-Face:** Dedicated buck regulator isolates pulse transients from the logic rail.

- **Logic/compute:** Buck to 5 V (CM5, camera); LDO to 3.3 V (MCU, sensors, radios).

Table 9: Battery comparison, PR1 vs. PR2 (80 % of energy usable).

	PR1: 2S 505080	**PR2: 3S Vapcell M11**
Voltage (nominal / full / cutoff)	7.4 / 8.4 / 6.6 V	11.1 / 12.6 / 9.0 V
Capacity / energy (usable)	3000 mAh / 22.2 Wh (17.8)	1100 mAh / 12.2 Wh (9.8)
Run time, Phase A–B (∼3.1 W)	5.8 h	**3.2 h**
Run time, Phase D (∼11.5 W)	1.5 h	**0.85 h**
Peak current (∼47 W)	7.1 A (2.4 C)	5.2 A (4.7 C, cell rated 10 A)
Motor PWM cap for 6 V	∼70 %	∼48 %
H-bridge on-time stall current	1.4 A/ch	2.1 A/ch

Table 10: Maximum load per rail.

Rail	**Max Current**	**Max Power**
Motor drive (PWM to 6 V)	4.0 A (6 V eq.); 2.1 A/ch at 12.6 V	24 W
5 V logic/compute	4.0 A	20 W
3.3 V (from 5 V LDO)	0.45 A	1.5 W
EP-Face + IR	TBD	TBD
Pack total (peak)	∼4.2 A @ 11.1 V; ∼5.2 A @ 9.0 V	∼47 W

Run time roughly halves, which is acceptable for lab testing in Phases A–C. The higher pack voltage sets new ratings for the power board: H-bridges for at least 2.1 A per channel and a supply above 12.6 V with margin, and buck regulators with a 9–12.6 V input range.

5.8 Next Steps

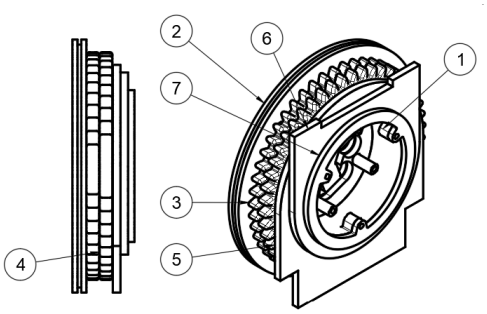
The tested Phase A-1 wiring and CubeMX configuration are carried over to design the Phase A main board and power board PCBs, with:

- H-bridges and buck regulators rated for the 3S pack (Section 5.7).
- The Phase A-1 pin map (Table 8) as the starting point for the main-board pin-out.

5.9 programming and debugging

6 Shaft Tolerance Analysis (∅55 mm Bearing)

The plain bearing (OD = 55 mm) acts as the shaft. All PETG-printed components have bores around it.



Parts List		
Item	Qty	Part Name
1	1	Wheel plain bearing v6
2	1	Wheel (left and right) v19
3	2	Spur gear v17
4	2	Shim Ring 55 68 0.2 v2
5	1	Lower-tilt-joint v12
6	1	Upper-tilt-joint v8
7	1	Bushing block v3

Figure 15: Shaft and bearing component arrangement.

Table 11: Joint classification and ISO fits for ∅55 mm bearing.

Joint	Component	Motion	Fit	ISO Class
J1	Wheel + 1st Spur Gear	Dynamic	Clearance	F7/h6
J2	2nd Spur Gear (Pan/Tilt)	Dynamic	Clearance	F7/h6
J3	Upper Tilt Joint	Dynamic	Clearance	F7/h6
J4	Lower Tilt Joint	**Static**	**Interference**	P7/h6

ISO 286 limits (bearing h6: 55.000–54.981 mm): Dynamic F7 bore 55.030–55.060 mm (clearance +0.030 to +0.079 mm); Static P7 bore 54.938–54.968 mm (interference +0.013 to +0.062 mm).

FDM print compensation: Printed holes come out undersized, so the ISO limits cannot be used directly as CAD dimensions. The analysis above was used as the starting point for a printed-fit test. PETG was chosen over nylon because it prints easily and the mechanical design is not a focus of this project.

Test result: A CAD bore of ∅55.00 + 0.15 = ∅55.15 mm was tested on the printed PETG parts and works. This single offset is adopted for **all** components on the shaft (wheel, spur gears, upper and lower tilt joints), replacing the separate dynamic and static bore sizes originally proposed (Table 12).

Table 12: Final CAD bore size (tested).

Part	Joint	CAD Bore (mm)
Wheel / Spur Gears / Upper Tilt	J1–J3	∅55.15
Lower Tilt Joint	J4	∅55.15

Bearing OD = ∅55.00 mm; offset +0.15 mm on all bores.

7 Teleoperation Architecture

Figure 16 shows the teleoperation stack. Controllers, topics and the robot model are shared between simulation and hardware; a launch argument selects the `ros2_control` hardware plugin.

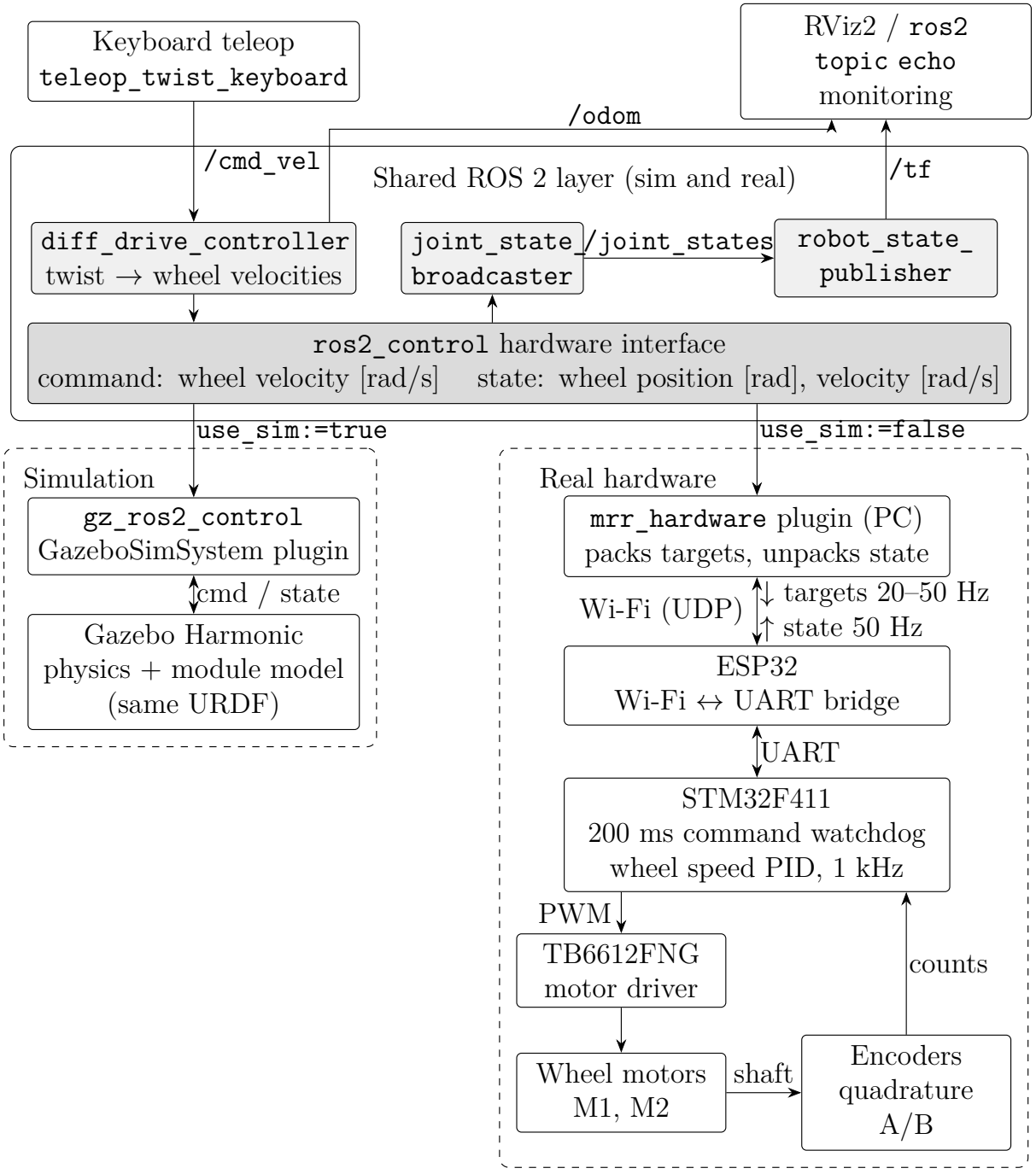


Figure 16: Teleoperation architecture. The controllers, topics and robot model are shared; a launch argument selects the hardware plugin under the `ros2_control` interface, either Gazebo through `gz_ros2_control` or one physical module over Wi-Fi.

8 Software Testing Plan (Phase A)

8.1 Objectives

Verify the integrated software pipeline: AprilTag detection and robot ID via external camera → real-time pose estimation → PID motor control → path planning and execution for MSRR configurations.

8.2 Setup

MSRR robot in a controlled indoor environment with an external camera, AprilTags for ID and pose reference, and a computing system running perception, localisation, planning, and control.

8.3 Goal Configurations

Snake, Car, DoubleDriver, L-shape.

8.4 Evaluation Metrics

- **Detection rate** — % frames with correct AprilTag ID.
- **Position / orientation error** — estimated vs. reference pose.
- **Tracking error** — desired vs. actual position and heading during path following.
- **Control response** — PID overshoot and steady-state error.
- **Processing rate** — perception + control update frequency.

8.5 Acceptance Criteria

Reliable tag detection, acceptable localisation and tracking error for real-time control, stable PID behaviour without excessive oscillation, and continuous operation without critical failures. Specific numerical thresholds to be defined based on system requirements.

8.6 Navigation and Avoidance Plan

This plan covers the **simulation stage** only.

Reconfiguration planning follows the parallel self-assembly method of Liu et al. [2], which was developed for SMORES-EP, the platform on which our module is based. The method consists of three stages: assigning each physical module to a position in the target configuration, determining the order in which modules connect, and driving each module into place to dock. Coordinated motion of multiple modules without collisions is handled by the prioritised route planner of Binder et al. [4, 5].

8.6.1 Planning Stages

A target configuration is represented as a tree in which each edge specifies the two connecting faces: T (top, the pan face), B (bottom), or L/R (the left and right wheel faces) [3]. The planar pose of each module is denoted $[x, y, \theta]$.

1. **Task assignment.** A root module is selected at the centre of the target configuration, and the pose of every target position is expressed relative to it. Likewise, the physical module closest to the centroid of the group is selected as the physical root, and all physical poses are expressed relative to it. The cost of assigning a module to a target position is the distance between their relative positions, and the Hungarian algorithm [6, 7] finds the assignment with the minimum total distance. The output is a goal pose for every module.
2. **Assembly sequence.** The configuration is assembled outward from the root, one tree layer at a time. Each module in a layer connects to a module that is already in place, so all modules in the same layer can move in parallel. Each connection is written $(m, c, \tilde{m}, \tilde{c}')$, meaning module m connects its face c to face $\tilde{c}'$ of module $\tilde{m}$. When several modules connect directly to the root, L/R connections are performed before T/B connections.
3. **Docking.** Each module navigates to the vicinity of its parent, aligns with the target face, and approaches it (Sections 8.6.2 and 8.6.3).

8.6.2 Navigation and Collision Avoidance

The workspace is discretised into a 10 cm grid [2], and routes are planned over its cells. Each module is modelled as a differential-drive robot, with forward velocity v and angular velocity ω produced by the two wheels:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 \\ \sin\theta & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v \\ \omega \end{bmatrix}. \tag{5}$$

Each module first navigates to a waypoint one grid cell away from its docking pose.

- **Prioritised planning.** Modules are assigned a priority order and planned sequentially. Each route is found with a time-aware A* search, in which a cell occupied by a previously planned module at a given time is treated as blocked. Waiting in place is permitted, and cell reservations overlap slightly in time so that two modules exchanging cells are detected as a collision.
- **Conflict resolution.** When a route is blocked, the planner applies five strategies in order: backtracking and waiting, yielding into a side branch at a crossing, yielding from the start cell, vacating the goal cell for a higher-priority module, and pushing the waiting position further along the route.
- **Rescheduling.** If no route is found, the maximum speed of the module causing the most conflicts is reduced. If this is insufficient, the priorities of the two most conflicting modules are swapped and planning is repeated.
- **Execution.** Each route segment carries a precondition of the form "module x has passed step n". Each module tracks its route with a PID controller and publishes its current step, allowing the other modules to check their preconditions.

8.6.3 Alignment and Approach

On reaching its waypoint, the module aligns with the target face. Let (x', y', θ') denote the module's pose error relative to its docking pose. For L/R connections, the controller

drives x' and θ' to zero; for T/B connections, it drives y' and θ' to zero. The module then moves straight forward until it contacts its parent. As the simulation does not model the EP magnets, a dock is considered successful when the modules are in contact and the residual error is within a set tolerance. On the physical system, L/R connections require a helper module to lift the moving module [2]; this connection type may therefore be excluded to reduce project complexity.

8.6.4 Test Procedure

Setup. Gazebo Fortress spawns N modules in a row, each with its own differential-drive controller. Ground-truth poses from the simulator replace AprilTag localisation, and the linear speed is limited to 0.049 m/s to match the physical motors.

Tests. Tests are performed in increasing order of complexity: the planner in isolation (verifying that the assignment is optimal against an exhaustive search and that the assembly sequence is correct); single-module navigation to a goal; two-module conflict scenarios (head-on, crossing paths, an occupied goal, and a narrow passage); alignment and approach; and finally assembly of the goal configurations listed above.

Metrics. Success rate, number of collisions (centre distance below one module width), planning time, total completion time, path-tracking error, and final alignment error.

Initial test case. The first planner test uses three modules connected side by side (v_3–v_2–v_1, joined L to R), starting from the default simulation spawn positions. The expected output is shown in Table 13: the middle module becomes the root, and both outer modules connect to it in parallel. As this configuration uses only L/R connections, it would be affected if that connection type is excluded.

Table 13: Expected planner output for three modules side by side, starting at $y = -0.3, 0, 0.3$ m.

Target position	**Assigned module**	**Goal pose** (x, y, θ)
v_2 (root)	module_2 (root)	$(0,\ 0,\ 0°)$
v_1	module_3	$(0,\ +0.1,\ 0°)$
v_3	module_1	$(0,\ -0.1,\ 0°)$
Parallel connections: (module_3, R, module_2, L), (module_1, L, module_2, R)		

9 Preparation for Phase B: EP Magnet

The docking system uses an Electro-Permanent Magnet (EPM) connector based on SMORES-EP [1]. Each EP unit pairs an AlNiCo 5 magnet (low coercivity, switchable) with an NdFeB magnet (high coercivity, fixed) between steel pole pieces, wound with a copper coil.

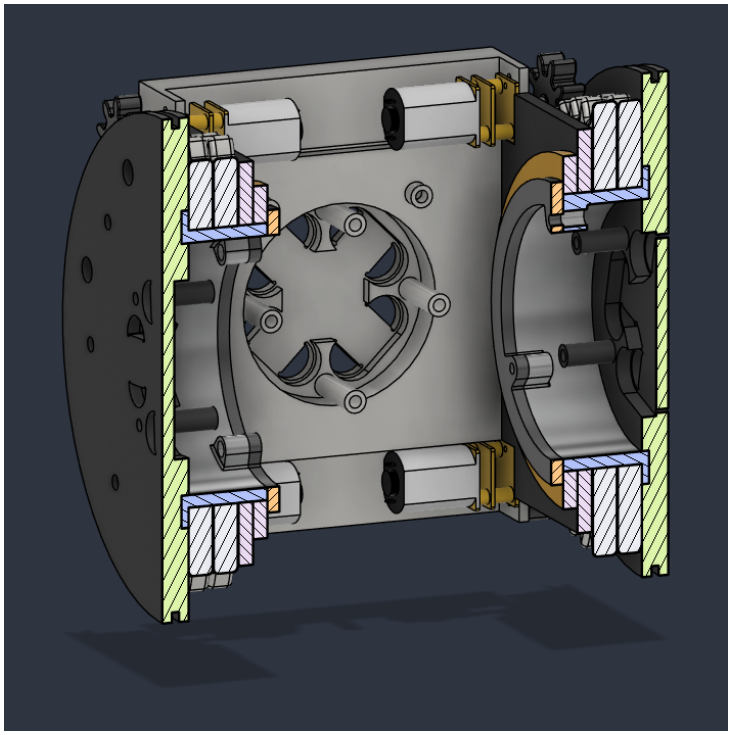


Figure 17: Interior view of the rectangular magnet housing.

Operating principle: A current pulse through the coil reverses AlNiCo 5 magnetisation while NdFeB stays fixed. Fields aligned → **ON** (attract); fields opposed → **OFF** (release). No power needed to hold either state.

The original EP-Face connector was developed for a significantly smaller robotic framework. To accommodate our larger target module dimensions of $100 \times 100 \times 100$ mm, we had to redesign the baseline architecture. Furthermore, due to the strict standard sizes of magnets available in the Thai market, we altered the core magnet dimensions and completely recalculated the system's electrical and mechanical properties to ensure compatibility and structural integrity.

9.1 System Architecture and Parametric Scaling

To fulfill the increased load requirements of our larger module, we scaled several primary parameters from the original reference design:

- **Magnet Dimensions:** The original design utilized 4.76×9.53 mm magnets. We increased this to 5×15 mm for both the high-coercivity NdFeB N52 and low-coercivity AlNiCo 5 cylinder magnets.
- **Module Integration:** The original magnets were arranged in a ∅42 mm ring. To fit our larger module face, the 4 EP magnets are now arranged within a ∅50 mm ring.

- **Physical Footprint:** The overall length of our EP magnet assembly is 25.6 mm. The pole piece body has a diameter of 10.6 mm, while the protruding contact foot has a diameter of 9.6 mm. The total height of the pole piece is 11.8 mm, with the contact plane extending 0.5 mm outward from the module face.
- **Weight:** The increased dimensions raise the weight from 10.6 g to 19.4 g per EP magnet.

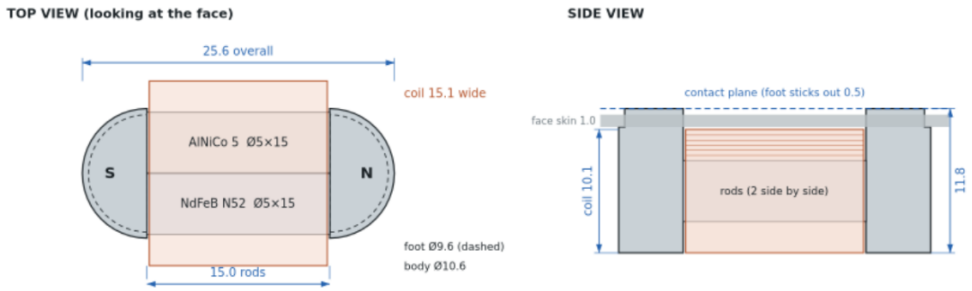


Figure 18: Top and side view dimensions of the redesigned 5×15 mm EP Magnet.

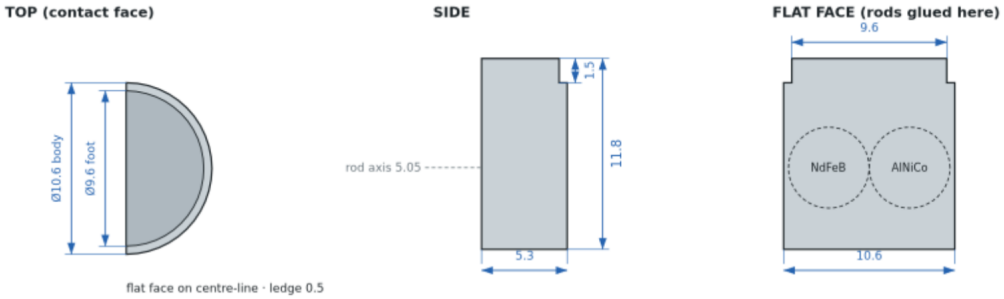


Figure 19: Detailed mechanical drawing of the low-carbon steel pole piece, illustrating the body and protruding contact foot.

9.2 Coil Design and Electrical Specifications

The coil winding was strictly recalculated to maximize the magnetic field without exceeding the physical constraints of the 11.8 mm pole pieces.

- **Wire Selection:** We transitioned from the original AWG 30 wire to SWG 28 double-coated enameled copper wire.
- **Winding Configuration:** The coil consists of exactly 210 turns. This is distributed precisely into 6 complete layers, with each layer containing exactly 35 turns.
- **Wire Length:** Due to the capsule shape of the coil wrapped around the two magnet bars, the perimeter expands with each subsequent layer. The first layer has a perimeter of 27.3 mm, expanding to 40.4 mm by the sixth layer. The total length of copper wire required per EP magnet is 7115.3 mm.

Wire Length per Layer (the perimeter of the capsule-shaped coil around two magnet bars):: P = 2d + π(d + 2δ)) = 2(5)+ π(5 + 2δ)

Layer	6 A: Turns	6 A: P (mm)	6 A: Length (mm)	6 A: Cumulative (mm)	Ø50-10A: Turns per Layer	Ø50-10A: P (mm)	Ø50-10A: Length (mm)	Ø50-10A: Cumulative (mm)
1	35	27.3	956.6	956.6	35	27.3	956.6	956.6
2	35	30.0	1048.3	2005.0	35	30.0	1048.3	2005.0
3	35	32.6	1140.0	3145.0	35	32.6	1140.0	3145.0
4	35	35.2	1231.7	4376.7	35	35.2	1231.7	4376.7
5	35	37.8	1323.4	5700.2	35	37.8	1323.4	5700.2
6	35	40.4	1415.1	7115.3	35	40.4	1415.1	7115.3
7	35	43.1	1506.8	8622.1	0	43.1	0.0	7115.3
8	35	45.7	1598.5	10220.7	0	45.7	0.0	7115.3
9	35	48.3	1690.3	11910.9	0	48.3	0.0	7115.3
10	5	50.9	254.6	12165.5	0	50.9	0.0	7115.3
11	0	53.5	0.0	12165.5	0	53.5	0.0	7115.3
12	0	56.2	0.0	12165.5	0	56.2	0.0	7115.3
13	0	58.8	0.0	12165.5	0	58.8	0.0	7115.3
14	0	61.4	0.0	12165.5	0	61.4	0.0	7115.3
15	0	64.0	0.0	12165.5	0	64.0	0.0	7115.3
Total	**320.0**			**12,165.5**	**210.0**			**7,115.3**
							Meters	7.115301344

Figure 20: Wire length calculations per layer demonstrating the 6-layer, 210-turn configuration.

9.2.1 Mathematical Modeling and Formula Derivations

To ensure the larger system meets the required strength characteristics, the following mathematical formulas were applied to derive our final design parameters:

- **Magnet Area and Total Flux:** The cross-sectional area of the cylindrical magnet rod (A_{rod}) is calculated first.

$$A_{rod} = \frac{\pi d^2}{4} = \frac{\pi(5)^2}{4} = 19.63 \text{ mm}^2$$

  The total magnetic flux (Φ) combines the remnant magnetization of both the NdFeB and AlNiCo magnets.

$$\Phi = A_{rod}(Br_{Nd} + Br_{Al}) = 19.63(1.455 + 1.265) = 53.4 \text{ T} \cdot \text{mm}^2$$

- **Minimum Foot Area:** To prevent magnetic saturation within the low-carbon steel, we assume a saturation point (B_{sat}) of 1.5 T to find the minimum required area (A_{min}).

$$A_{min} = \frac{\Phi}{B_{sat}} = \frac{53.4}{1.5} = 35.6 \text{ mm}^2$$

  This minimum area dictates that the minimum foot diameter (D_{min}) must be 9.52 mm.

$$D_{min} = \sqrt{\frac{8A_{min}}{\pi}} = 9.52 \text{ mm}$$

- **Magnetic Flux Density in Foot:** By utilizing our selected foot diameter of 9.6 mm, the actual surface area is 36.2 mm^2. This yields an actual magnetic flux density (B) at the contact surface.

$$B = \frac{\Phi}{A_{foot}} = \frac{53.4}{36.2} = 1.48 \text{ T}$$

- **Holding Force Calibration:** Magnetic holding force is proportional to $B^2A/(2\mu_0)$. We scale the force directly from the reference paper's maximum baseline force of 90 N (F_{paper}). The scaling multiplier 1.078 represents the proportional increase in

strength due to our larger contact area and stronger flux. The term $f(H)$ acts as a compensation factor for polarization reversal percentage.

$$F = F_{paper} \times \frac{(B^2A)}{(B^2A)_{paper}} \times f(H) = 90 \text{ N} \times 1.078 \times f(H)$$

- **Mean Length of Turn (MLT):** Because the coil is wrapped around two parallel magnet bars, the winding forms a capsule shape. The length for each coil layer j includes the two straight sides ($2d$) and the two curved ends ($\pi(d + 2\delta_j)$), where δ_j represents thickness increment per layer.

$$MLT_j = 2d + \pi(d + 2\delta_j)$$

- **Coil Resistance:** The total resistance is computed using the resistivity of copper (ρ) and the sum of the wire lengths across all layers.

$$R = \frac{\rho \sum(N_j \cdot MLT_j)}{A_{wire}} = \frac{1.724 \times 10^{-8} \times 7.12}{0.111 \times 10^{-6}} = 1.105 \ \Omega$$

- **Current and Magnetic Field (H):** Operating at 12 V, we estimate an external resistance (R_{ext}) of 0.15 Ω for the wiring and MOSFET components.

$$I = \frac{V}{(R + R_{ext})} = \frac{12}{1.255} = 9.56 \text{ A}$$

This current generates an applied field capable of fully reversing the AlNiCo 5 polarity.

$$H = \frac{NI}{L} = \frac{2008}{0.015} = 134 \text{ kA/m}$$

- **Switching Energy:** The total energy consumed to switch states is calculated by multiplying voltage, current, duration per pulse (4 ms), and number of pulses (3).

$$E = V \cdot I \cdot t \cdot 3 = 12 \times 9.56 \times 0.004 \times 3 = 1.38 \text{ J}$$

9.2.2 Performance Metrics

The new coil and magnet configuration yields significant performance improvements over the original paper:

- **Electrical Resistance:** Our configuration yields a coil resistance of 1.105 Ω, compared to the original 2.08 Ω.
- **Magnetic Field and Current:** Driven at 12 V, the system draws 9.56 A, generating an operating field of 134 kA/m. At a peak battery voltage of 12.6 V, the current reaches 10.04 A, generating a field of 141 kA/m. The original design achieved ≈ 6 A and 121 kA/m.
- **Holding Force:** The contact surface area is 36.2 mm^2 with a magnetic flux density of ≈ 1.48 T. This results in an expected holding force of ≈ 32 N per EP pair, yielding a total face holding force of ≈ 97 N. The original paper averaged 28.3 N per pair and 88.4 N per face.
- **Switching Energy:** The required switching energy per face is ≈ 5.5 J, utilizing three 4 ms pulses per EP. The original system required 2.5 J using 3 ms pulses.

9.3 Current Progress and Procurement Challenges

All theoretical calculations, parameter definitions, and geometric models have been successfully completed. We have ordered the SWG 28 double-coated enameled copper wire, NdFeB N52 magnets, Kapton tape, and high-viscosity CA glue gel.

However, we are navigating two primary manufacturing and procurement hurdles before constructing our initial 8-EP prototype:

1. **AlNiCo 5 Sourcing:** The primary supplier is in China, creating purchasing and logistical roadblocks. It is exceptionally hard to find the perfect size ($\varnothing 5 \times 15$ mm) and grade (AlNiCo 5). We are actively seeking alternative vendors to prevent assembly delays.
2. **Pole Piece Fabrication:** Standard external machine shops and wire-cut services refuse to accept our custom cutting requirements due to the low-carbon steel pole pieces being exceptionally small. We plan to utilize the CNC machining facilities at the university to manufacture these pieces ourselves.

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